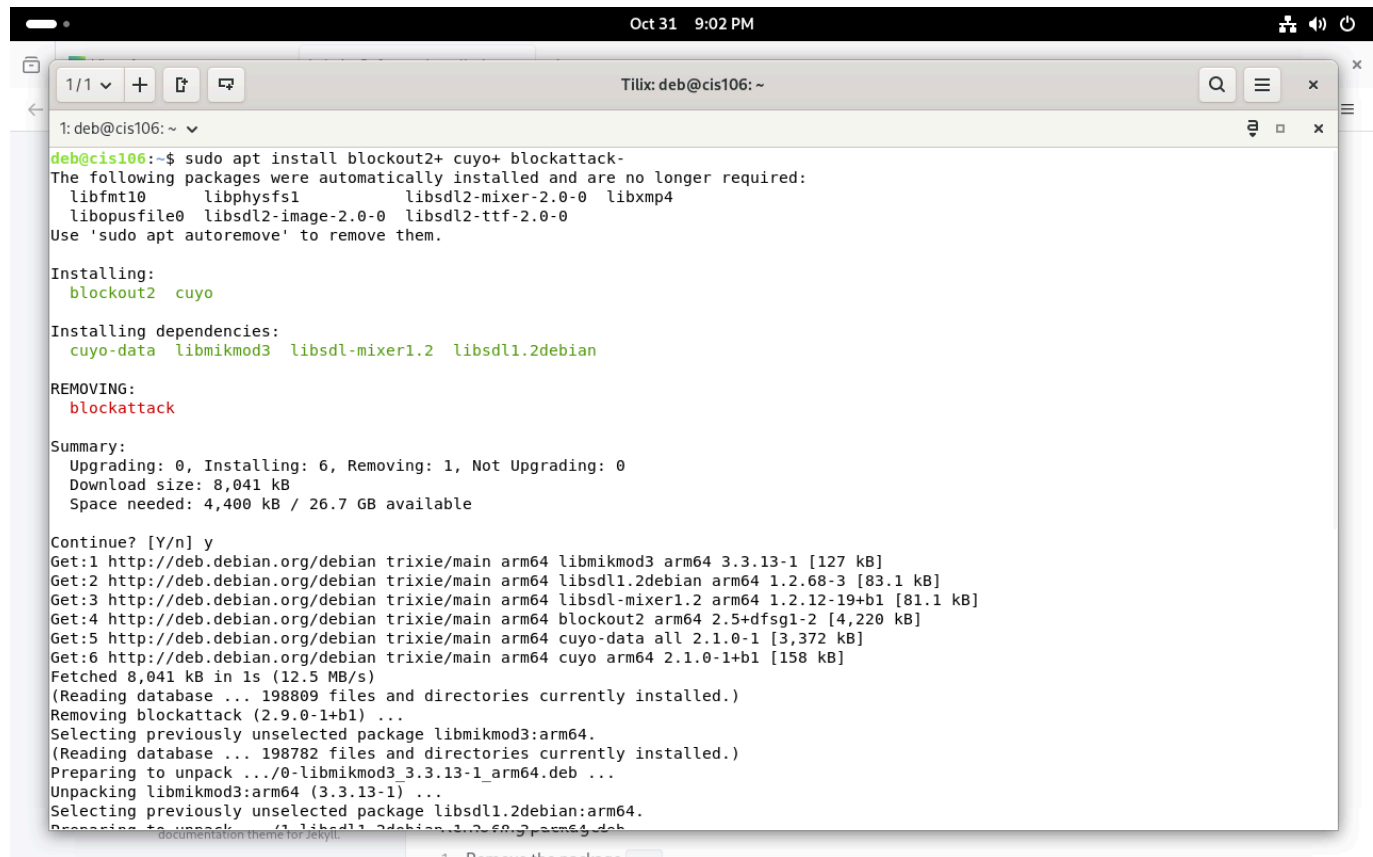


Question 1



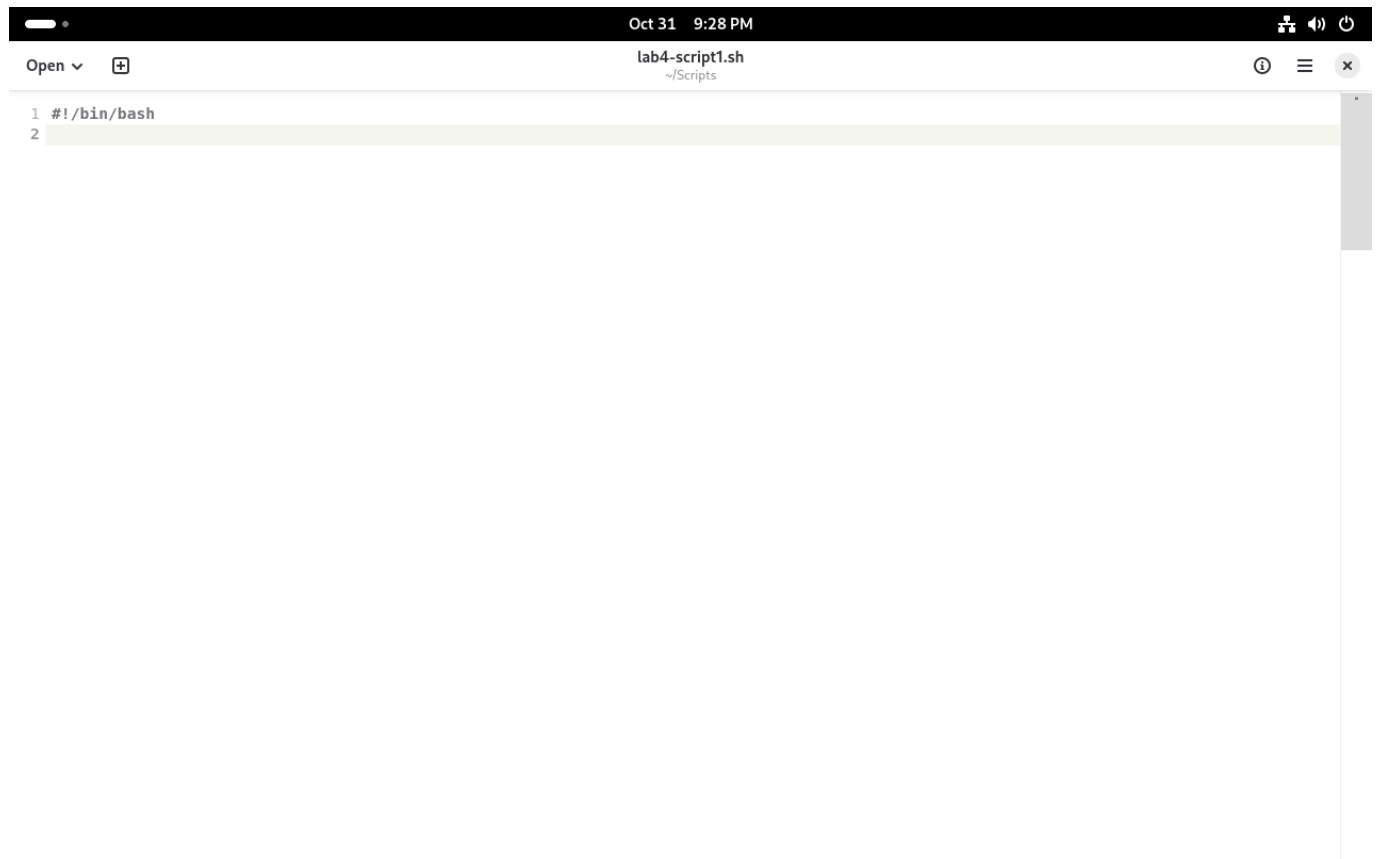


The image shows two screenshots of a terminal window on a system named 'deb@cis106'. The terminal window has a title bar with 'Tilix: deb@cis106: ~' and standard window controls. The first screenshot shows the command 'sudo apt remove cuyo' being executed. The output indicates that several packages were automatically installed and are no longer required: 'cuyo-data', 'libfmt10', 'libopusfile0', 'libphysfs1', 'libsdl2-image-2.0-0', 'libsdl2-mixer-2.0-0', 'libsdl2-ttf-2.0-0', and 'libxmp4'. The terminal shows the removal of 'cuyo' and a summary of the operation, including the space freed (562 kB). The second screenshot shows the command 'sudo apt purge cuyo' being executed. The output shows the removal of 'cuyo*' and a summary of the operation, including the space freed (7,631 kB). The terminal window also shows the command 'sudo apt clean && sudo apt autoclean && sudo apt autoremove && sudo apt autopurge' being executed, which removes the specified packages and their dependencies.

```
1: deb@cis106: ~  
deb@cis106:~$ sudo apt remove cuyo  
The following packages were automatically installed and are no longer required:  
  cuyo-data libfmt10 libopusfile0 libphysfs1 libsdl2-image-2.0-0 libsdl2-mixer-2.0-0 libsdl2-ttf-2.0-0 libxmp4  
Use 'sudo apt autoremove' to remove them.  
  
REMOVING:  
  cuyo  
  
Summary:  
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 0  
  Freed space: 562 kB  
  
Continue? [Y/n] y  
(Reading database ... 199880 files and directories currently installed.)  
Removing cuyo (2.1.0-1+b1) ...  
Processing triggers for hicolor-icon-theme (0.18-2) ...  
Processing triggers for gnome-menus (3.36.0-3) ...  
Processing triggers for mailcap (3.74) ...  
Processing triggers for desktop-file-utils (0.28-1) ...  
deb@cis106:~$ sudo apt purge cuyo  
The following packages were automatically installed and are no longer required:  
  cuyo-data libfmt10 libopusfile0 libphysfs1 libsdl2-image-2.0-0 libsdl2-mixer-2.0-0 libsdl2-ttf-2.0-0 libxmp4  
Use 'sudo apt autoremove' to remove them.  
  
REMOVING:  
  cuyo*  
  
Summary:  
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 0  
  Space needed: 0 B / 26.7 GB available  
  
Continue? [Y/n] y  
(Reading database ... 199871 files and directories currently installed.)  
Purging configuration files for cuyo (2.1.0-1+b1) ...  
deb@cis106:~$  
documentation theme for Jekyll.  
1. Remove the package 'cuyo'
```

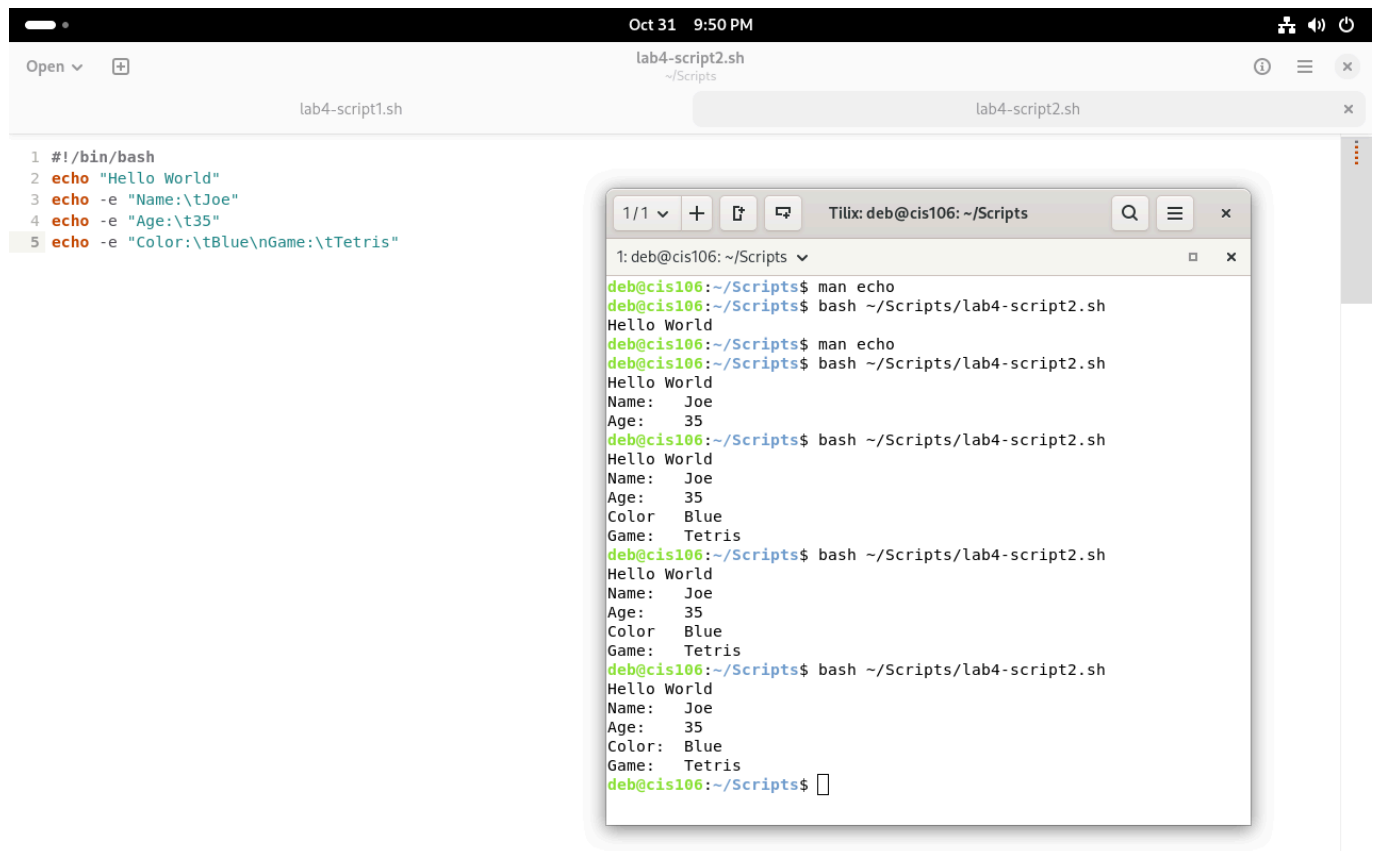
```
1: deb@cis106: ~  
deb@cis106:~$ sudo apt clean && sudo apt autoclean && sudo apt autoremove && sudo apt autopurge  
REMOVING:  
  cuyo-data libfmt10 libopusfile0 libphysfs1 libsdl2-image-2.0-0 libsdl2-mixer-2.0-0 libsdl2-ttf-2.0-0 libxmp4  
  
Summary:  
  Upgrading: 0, Installing: 0, Removing: 8, Not Upgrading: 0  
  Freed space: 7,631 kB  
  
Continue? [Y/n] y  
(Reading database ... 199871 files and directories currently installed.)  
Removing cuyo-data (2.1.0-1) ...  
Removing libfmt10:arm64 (10.1.1+ds1-4) ...  
Removing libsdl2-mixer-2.0-0:arm64 (2.8.1+dfsg-2) ...  
Removing libopusfile0:arm64 (0.12-4+b3) ...  
Removing libphysfs1:arm64 (3.0.2-6+b2) ...  
Removing libsdl2-image-2.0-0:arm64 (2.8.0+dfsg-1) ...  
Removing libsdl2-ttf-2.0-0:arm64 (2.24.0+dfsg-2) ...  
Removing libxmp4:arm64 (4.6.2-2) ...  
Processing triggers for man-db (2.13.1-1) ...  
Processing triggers for libc-bin (2.41-12) ...  
Summary:  
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0  
deb@cis106:~$  
documentation theme for Jekyll.  
Therefore, to create a shell script, all you need is a text editor. In your virtual machine, you have 3 text editors installed:
```

Question 2



The screenshot shows a terminal window with a dark theme. The title bar indicates the date and time as "Oct 31 9:28 PM". The window title is "lab4-script1.sh" with a subtitle "~ /Scripts". The terminal content shows a script with two lines: "1 #!/bin/bash" and "2".

```
1 #!/bin/bash
2
```

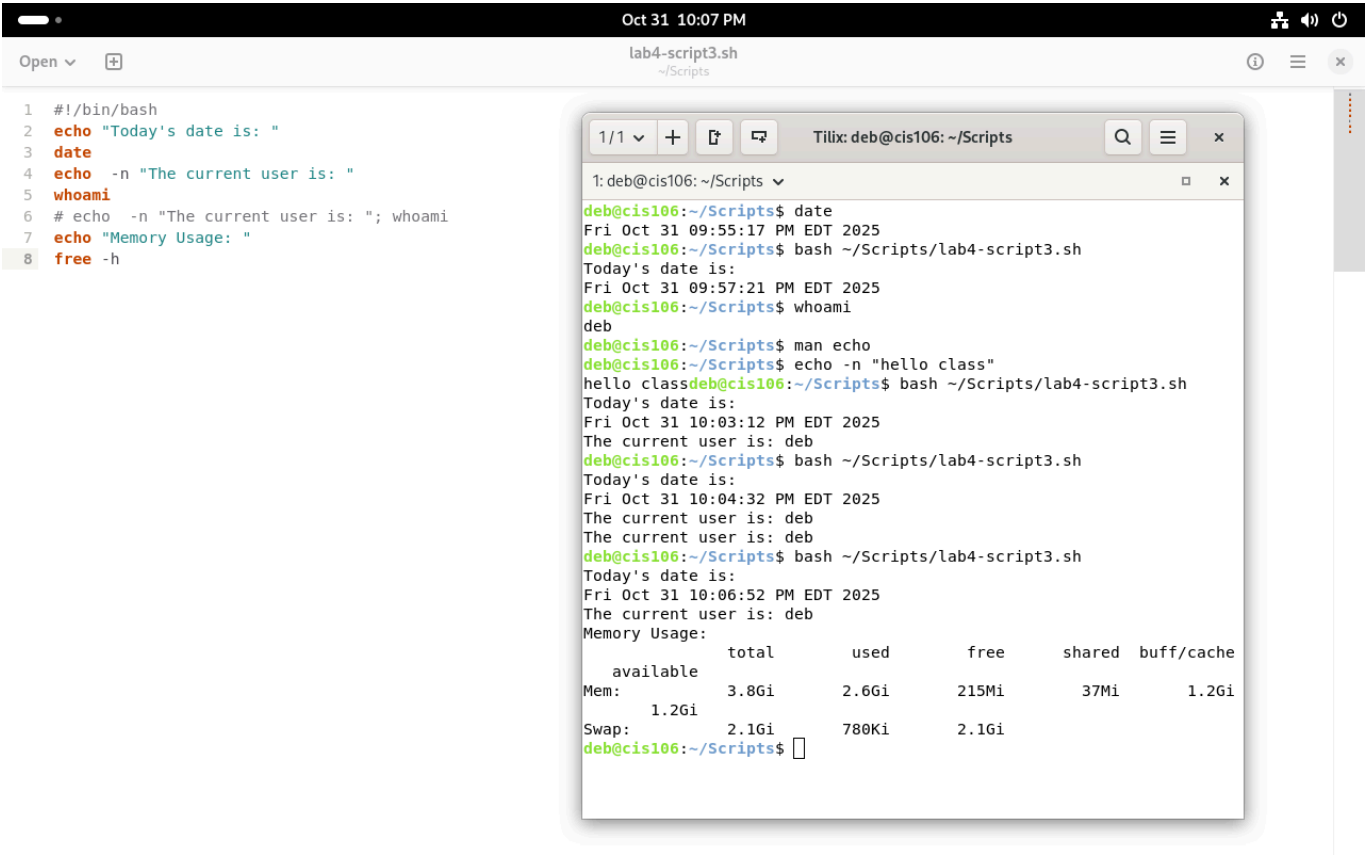


The screenshot shows a terminal window with a dark theme. The title bar indicates the date and time as "Oct 31 9:50 PM". The window title is "lab4-script2.sh" with a subtitle "~ /Scripts". The terminal content shows a script with five lines: "1 #!/bin/bash", "2 echo \"Hello World\"", "3 echo -e \"Name:\\tJoe\"", "4 echo -e \"Age:\\t35\"", and "5 echo -e \"Color:\\tBlue\\nGame:\\tTetris\"".

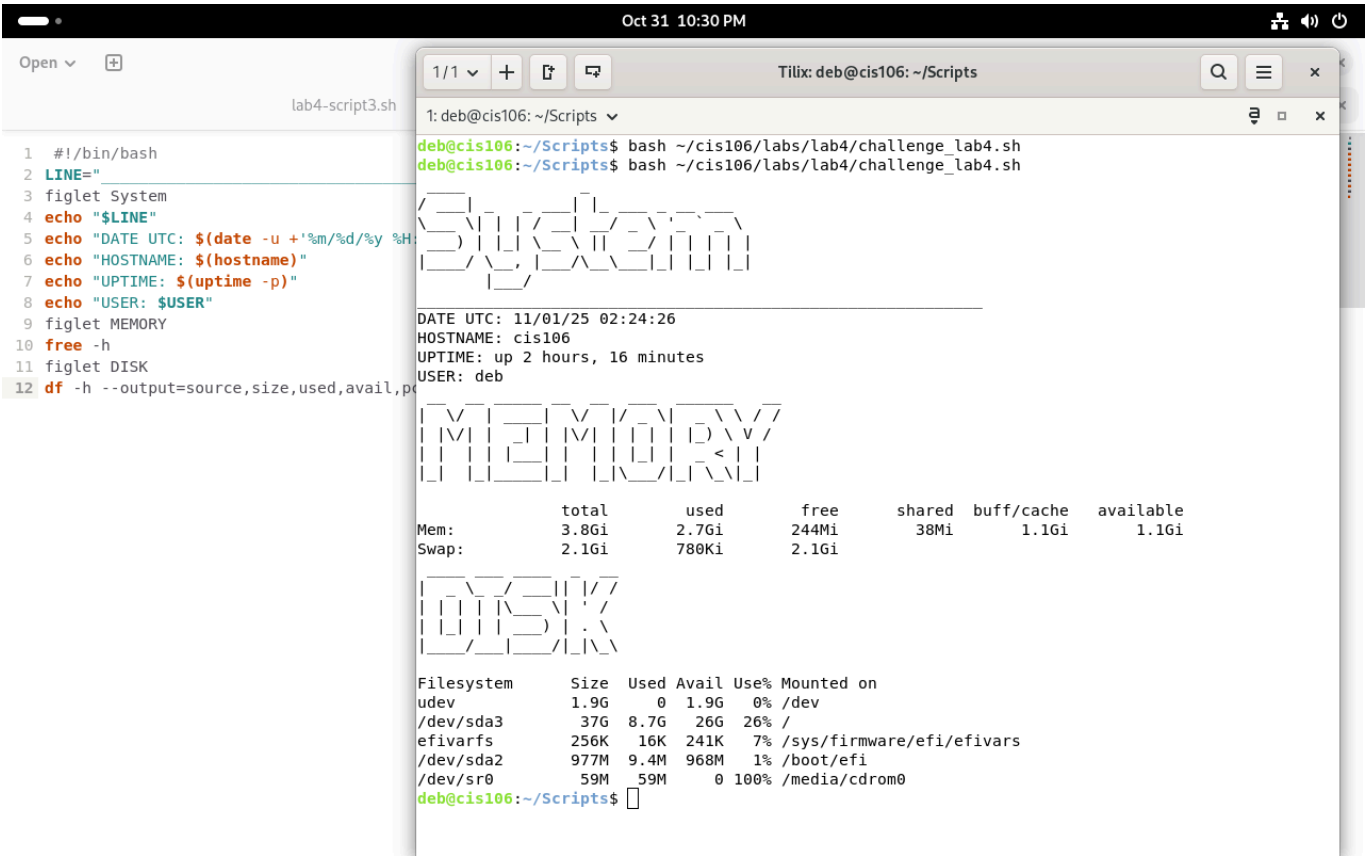
```
1 #!/bin/bash
2 echo "Hello World"
3 echo -e "Name:\\tJoe"
4 echo -e "Age:\\t35"
5 echo -e "Color:\\tBlue\\nGame:\\tTetris"
```

Below the terminal window, there is a smaller window titled "Tilix: deb@cis106: ~/Scripts" showing the execution of the script multiple times. The output of the script is displayed in the terminal window.

```
1: deb@cis106: ~/Scripts
deb@cis106:~/Scripts$ man echo
deb@cis106:~/Scripts$ bash ~/Scripts/lab4-script2.sh
Hello World
deb@cis106:~/Scripts$ man echo
deb@cis106:~/Scripts$ bash ~/Scripts/lab4-script2.sh
Hello World
Name: Joe
Age: 35
deb@cis106:~/Scripts$ bash ~/Scripts/lab4-script2.sh
Hello World
Name: Joe
Age: 35
Color: Blue
Game: Tetris
deb@cis106:~/Scripts$ bash ~/Scripts/lab4-script2.sh
Hello World
Name: Joe
Age: 35
Color: Blue
Game: Tetris
deb@cis106:~/Scripts$ bash ~/Scripts/lab4-script2.sh
Hello World
Name: Joe
Age: 35
Color: Blue
Game: Tetris
deb@cis106:~/Scripts$
```



Question 3



Scripts